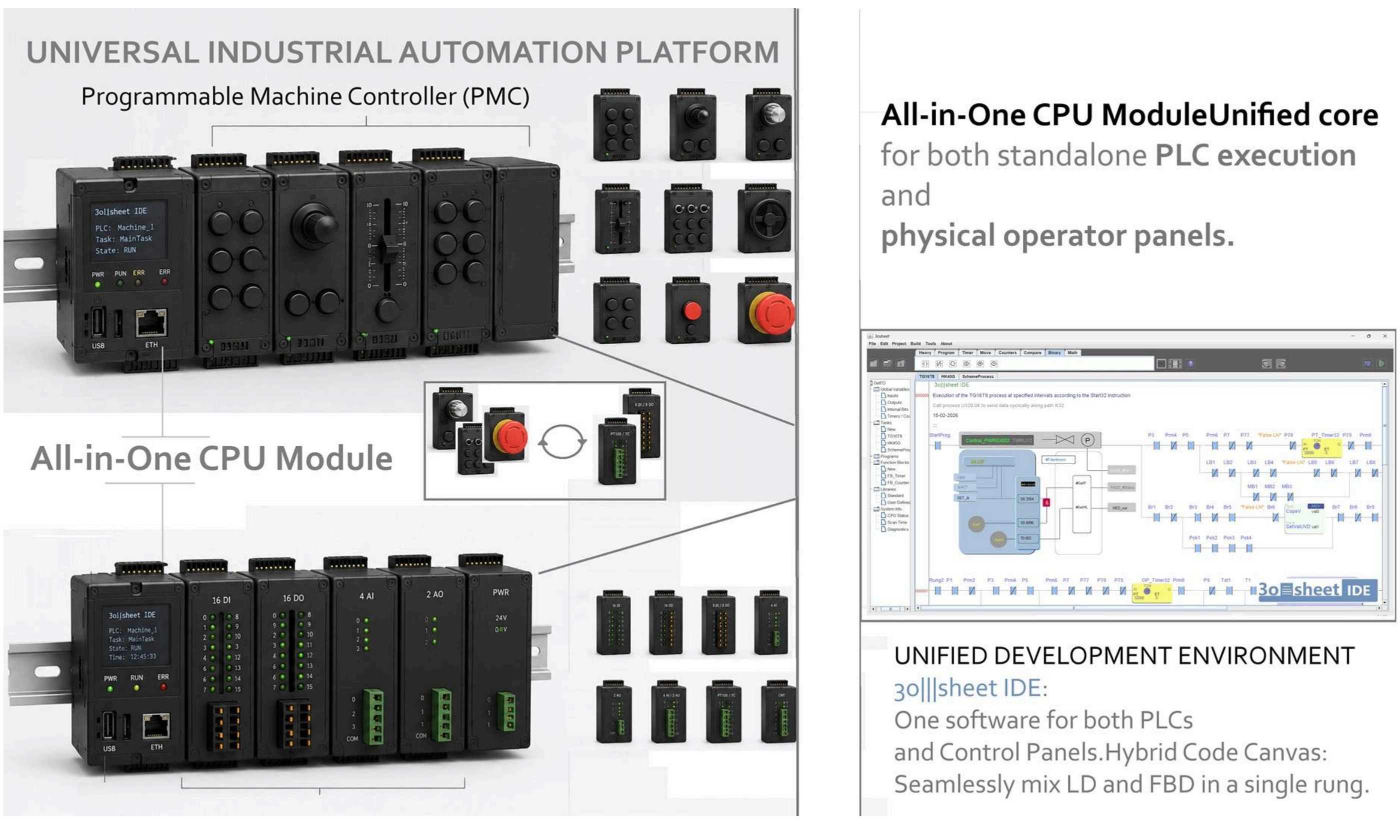
Programmable Machine Controller. PMC

Technical Specification: High-Ergonomics GARM Architecture & Ecosystem



1. System Uniqueness & Paradigm Shift

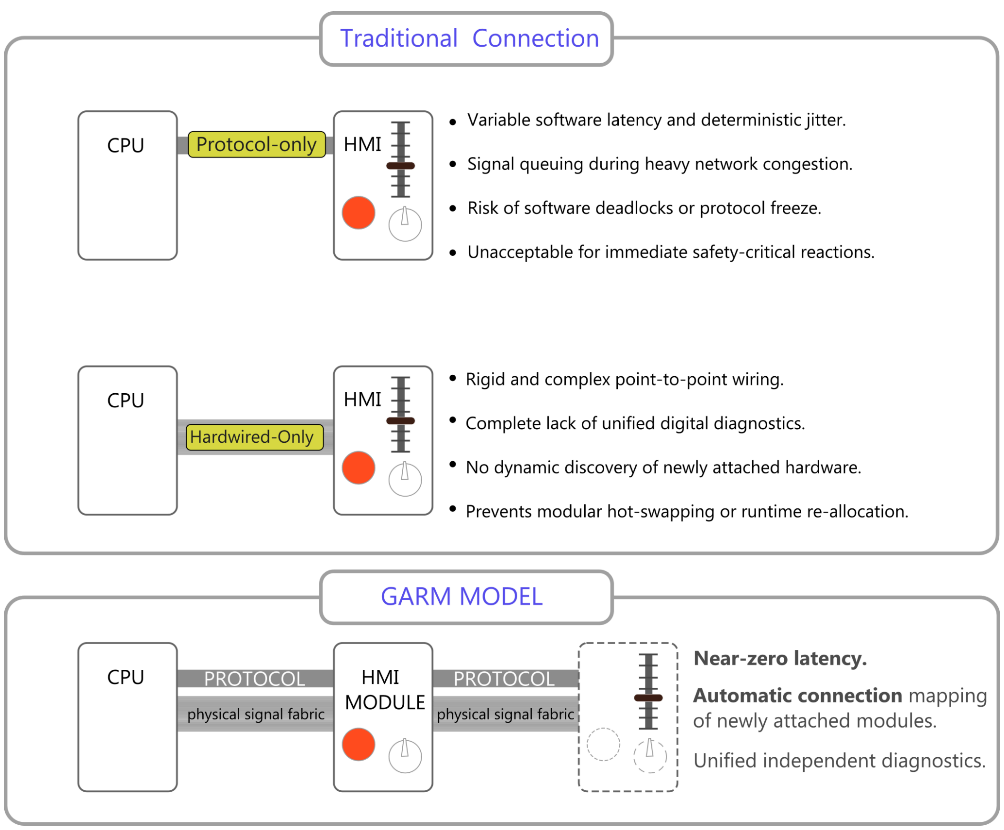
The Programmable Machine Controller (PMC) represents an unprecedented, industry-first hardware-software ecosystem that completely redefines traditional industrial automation interfaces. Traditional setups strictly segregate machine control logic (the PLC) from operator terminal interfaces (HMI panels) and standard peripheral I/O cards, forcing them to exchange data over slow, cyclic network communication layers. This separation causes significant software latency, data packet arbitration, and deterministic processing jitter.

The PMC breaks this paradigm through its proprietary **Unified Core Architecture**. It seamlessly merges the industrial computing core, standard field expansion I/O, and physical high-ergonomics operator controls (buttons, analog joysticks, rotary encoders, and switches) into a single, unified execution and mechanical layer on a standard DIN rail. By eliminating intermediate communication barriers, the controller achieves complete architectural fusion.

2. The Core Mechanics of the GARM Paradigm

To coordinate execution-critical physical controls alongside standard digital data traffic on a shared interface, the platform incorporates the **GARM** structural principle. GARM explicitly splits real-time control mechanics into two autonomous, parallel operational planes that interact asynchronously without blocking one another:

- **The Physical Execution Plane:** When an operator triggers a hardware input component (e.g., a physical button or joystick), the action immediately directs a raw electrical potential down a dedicated line within the physical fabric. This signal routes straight to the central controller processor core or actuator inhibitor circuit, entirely bypassing software protocol stacks, packetization overhead, and buffering queues.
- **The Digital Verification Plane:** Concurrently and independently, a parallel digital protocol channel packages the event information into a standard communication packet. This digital track handles capability logging, resource mapping, global diagnostics, and asynchronous software status confirmation post-factum within a deterministic 100 to 400 microseconds heartbeat interval.



3. Hybrid Inter-Module Interface & Zero-Protocol Mapping

The unified physical backbone comprises a specialized shared signal fabric containing a fixed pass-through bundle of 36 to 72 dedicated raw conductors alongside an independent digital protocol line. When an ergonomic operator block or standard high-density I/O module docks onto the common rail system, it initiates an automated **Zero-Protocol Mapping** handshake:

1. **Discovery & Interrogation:** The newly attached hardware component transmits its digital profile packet through the digital plane to the CPU. This profile explicitly defines its exact operational hardware layout (e.g., specific counts of push-buttons, rotary dials, or analog joysticks).
2. **Dynamic Commutation:** The central processor analyzes its current physical resource grid, isolates free lines from the 36-72 hardware pool, and sends back an allocation map. The module instantly reconfigures its internal solid-state switching matrix to map the physical controls directly onto the designated lines of the shared hardware bus.
3. **IDE Variable Synchronization:** Within the **unified 3o|||sheet engineering environment**, all physical and touchscreen interface elements map directly to abstract logical identifiers and internal controller variables. The hardware level synchronization entirely avoids manual network address allocation, variable linking, or packet decoding blocks inside the controller application software.

4. Runtime Criticality Tiers & Resource Allocation

To safeguard system stability, optimize conductor utilization, and enforce deterministic timing bounds, the PMC system strictly classifies all incoming runtime signals into three dedicated architectural tiers:

Execution Tier	Physical Bus Routing	Verification Strategy
Critical Execution Tier (e.g., E-Stops, Safety Interventions)	Direct hardware potential matrix via the 36-72 physical conductor lines.	Asynchronous 10-100 ms status heartbeat over digital channel for post-factum logging.
Important Execution Tier (e.g., Operational Joysticks, Dials)	Direct hardware potential routing straight into core CPU terminal registers.	Processing status feedback loop returned via digital channel within execution scan.
Ordinary Execution Tier (e.g., Thermocouples, Backlights)	Exclusively packetized and transmitted via serial digital bus; consumes 0 wire resources.	Fully encapsulated synchronous or asynchronous communication protocol transactions.